

Hedging with Event Contracts: A System for Peer-to-Peer Risk Transfer

Vardhan Shorewala, Joonbeom Kwon, Jatin Singaraju
polyhedg.com

Abstract. A purely peer-to-peer version of risk transfer would allow economic losses contingent on specific real-world events to be offset without relying on a traditional financial intermediary like an insurance company. Existing derivatives provide part of the solution for price-based risks, but the main benefits are lost if a trusted third party is still required to underwrite and price discrete, non-financial event risks. We propose a solution to the problem of unhedgeable event risk using a peer-to-peer network. The network prices contracts by creating a market for a binary outcome, forming a public record of the consensus probability that cannot be changed without significant capital expenditure. The market price not only serves as proof of the consensus view but also as the mechanism for transferring risk. As long as a majority of capital seeks to price risk accurately, it will generate a fair price and absorb hedging demand. The network itself requires minimal structure. Buy and sell orders are broadcast on a best effort basis, and participants can leave and rejoin the network at will, accepting the current market price as proof of the collective consensus on the event’s probability.

1. Introduction

Commerce has come to rely almost exclusively on financial institutions serving as trusted third parties to manage risk. While the system works well enough for standard price volatility (e.g., futures, options), it still suffers from the inherent weaknesses of the trust-based model for discrete, non-financial events. Hedging a regulatory decision, an election outcome, or a geopolitical event is not truly possible, since financial institutions cannot create liquid, specific markets for such outcomes. The cost of bespoke insurance is prohibitive, and there is a broader cost in the loss of ability for businesses to make confident decisions in the face of uncertainty. With the possibility of a large, unmitigated loss, the need for caution spreads. Businesses must be wary of future events, paralyzing them with analysis and preventing long-term capital allocation. A certain amount of event risk is accepted as unavoidable.

What is needed is a risk transfer system based on market-based proof instead of trust, allowing any two willing parties to transact risk directly with each other without the need for a trusted insurer. Contracts that are computationally impractical for a single party to manipulate would protect hedgers from mispriced risk, and routine market-making mechanisms could easily be implemented to provide liquidity. In this paper, we propose a solution to the event-risk problem using a peer-to-peer distributed market to generate computational proof of an event’s probability, which in turn facilitates a direct hedge. The system is secure as long as rational, informed actors collectively contribute more capital than any cooperating group of malicious or uninformed actors.

2. Event Contracts

We define an event contract as a digital instrument whose final value is determined by the outcome of a specific, real-world event. Each contract is tied to a binary resolution: it pays out a fixed amount (e.g., \$1) if the event occurs (“Yes”) and \$0 otherwise (“No”). A hedger can verify the contract’s terms by inspecting its publicly defined resolution criteria.

To hedge a potential loss, a party acquires contracts corresponding to the outcome they wish to insure against.

The problem, of course, is determining a fair price for this contract. A common solution is to introduce a trusted central authority, or bookmaker, that sets odds. The problem with this solution is that the fate of the entire risk transfer system depends on the entity setting the price. We need a way for the hedger to know that the price reflects the true, consensus-based probability of the event. To accomplish this without a trusted party, the price discovery must be publicly announced, and we need a system for participants to agree on a single price at the time of the transaction. The hedger needs proof that at the time of their trade, the majority of capital agreed it was a fair price.

3. The Market Mechanism

The solution we propose begins with a market mechanism. A market works by aggregating the beliefs of many participants into a single price. The price of a contract that pays \$1 on an outcome is the market’s estimate of the probability of that outcome. This price is published widely through an order book or an Automated Market Maker (AMM). The price proves that the market’s collective wisdom, at that moment, valued the probability at that level. Each new trade reinforces or adjusts this price.

4. Proof-of-Capital

To implement a distributed market on a peer-to-peer basis, we use a proof-of-capital system. Rather than one-IP-address-one-vote, which could be subverted by anyone able to allocate many IPs, proof-of-capital is essentially one-dollar-one-vote. The consensus probability is represented by the price that has the most capital supporting it. If a majority of capital is controlled by honest nodes (participants seeking to accurately price the event), the market price will grow to be the most accurate forecast and outpace any attempts at manipulation. To modify the price, a manipulator would have to expend enormous capital buying or selling against the honest participants. We will show later that the probability of a manipulator sustaining an inaccurate price diminishes exponentially as more rational capital enters the market.

5. Network

The steps to run the network for a hedging transaction are as follows:

1. A new event contract is created and broadcast to all nodes.
2. Each node (participant) can place buy (“Yes”) or sell (“No”) orders at prices reflecting their own belief. These orders are collected into a public order book.
3. The market-clearing price p is established.
4. An entity identifies a financial exposure of loss L that occurs if the event resolves to “Yes”.
5. To hedge this exposure, the entity broadcasts a buy order for L “Yes” contracts at the market price p .
6. The order is matched with sell orders from other nodes. The transaction, at a total cost of $L \bullet p$, is complete.
7. Upon the event’s resolution, if it is “Yes”, the L contracts pay out \$1 each, for a total of \$ L , offsetting the operational loss. If “No”, the contracts expire worthless.

The network always considers the market-clearing price to be the correct one. If two participants broadcast conflicting large orders simultaneously, some nodes may see one or the other first. In that case, the market works on the first one it received, but the price will quickly adjust as the second order propagates. The tie will be broken when liquidity fills one side of the order book, and a new stable price is found.

6. Incentive

By convention, the transaction to hedge a risk provides an incentive for other participants to take the other side of the trade. This adds an incentive for nodes (speculators or liquidity providers) to support the network. This provides a way to initially absorb risk into the system, since there is no central authority to underwrite it. The steady addition of capital from speculators is analogous to insurers deploying capital to underwrite policies. In our case, it is risk capital and information that is expended.

The incentive is funded by the price of the contract. If a speculator believes the true probability of an event is lower than the market price p , they can sell contracts, collecting a premium they believe is larger than the risk-warranted price. Once the risk is transferred, the incentive for the hedger is the removal of uncertainty, while the incentive for the speculator is the potential for profit.

7. Calculations

We consider the scenario of an entity transforming an uncertain, binary risk into a fixed, certain cost. The race between the two outcomes can be characterized as a simple probability problem.

Let L = the potential loss in dollars if the event occurs. Let p = the market price of a contract that pays \$1 if the event occurs. Let Q = the quantity of contracts purchased for the hedge.

To fully hedge the loss L , the entity must set $Q = L$.

The cost of this hedge is a fixed, upfront expense: **Cost of Hedge** = $Q \bullet p$
 $= L \bullet p$

We now evaluate the net financial outcome in both possible states of the world.

Scenario 1: The adverse event occurs. * Operational Impact: $-L$ * Hedge Payout: $+L$ (from the L contracts each paying out \$1) * Initial Cost of Hedge: $-(L \bullet p)$ * **Net Financial Result** = $-L + L - (L \bullet p) = -(L \bullet p)$

Scenario 2: The adverse event does not occur. * Operational Impact: \$0 * Hedge Payout: \$0 (from the L contracts expiring worthless) * Initial Cost of Hedge: $-(L \bullet p)$ * **Net Financial Result** = $\$0 + \$0 - (L \bullet p) = -(L \bullet p)$

As such, the verification is reliable. The entity has successfully transformed a catastrophic, uncertain risk of a $-L$ loss into a manageable and predictable expense of $-(L \bullet p)$. The binary risk is eliminated and replaced by a fixed, known, budgetable cost, which is the premium paid for certainty. The system allows a user to make the future's financial impact irrelevant.

Outcome	Operational Impact	Hedge Payout	Cost of Hedge	Net Result
Event Occurs	$-L +L$	$-L p * \textcircled{L \bullet p}^{**}$		
Event Does Not Occur	\$0	$0 \textcircled{L \bullet p}$	$-\$L \bullet p$	

8. Conclusion

We have proposed a system for peer-to-peer risk transfer without relying on trust. We started with the usual framework of insurance, which provides financial protection but is incomplete without a way to price and transfer risk for discrete, non-financial events in a decentralized manner. To solve this, we proposed a peer-to-peer network using proof-of-capital to record a public price for event contracts that quickly becomes computationally impractical for a single actor to manipulate if the market is sufficiently liquid. The network is robust in its unstructured simplicity. Nodes act all at once with little coordination. They do not need to be identified, since orders are not routed to any particular place and only need to be broadcast to the market. Nodes can leave and rejoin the network at will, accepting the market price as proof of the consensus probability. They vote with their capital, expressing their acceptance of the price by transacting at it and rejecting other prices by refusing to transact. Any needed rules and incentives can be enforced with this consensus mechanism.